# Architectural Fidelity

## Data Representation & Hardware Performance

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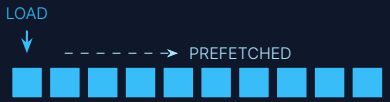
# The Mechanism: Layout determines Latency.

## Contiguous Memory

Hardware prefetcher's ideal case.

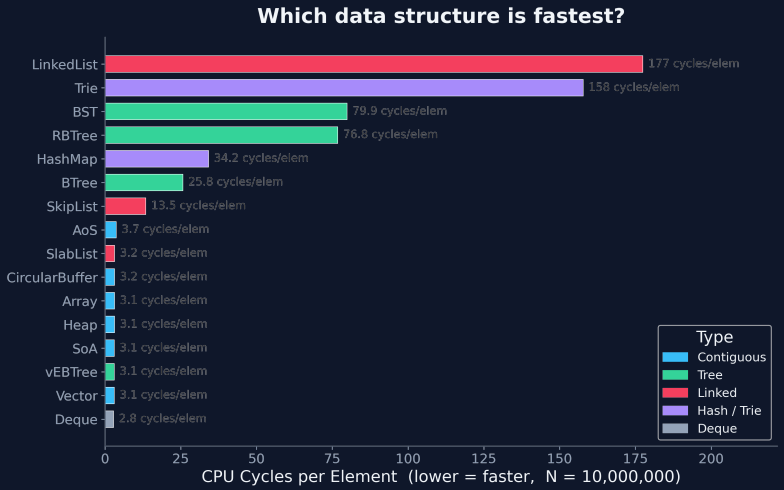



## Disjoint Memory

Inducing pipeline stalls.

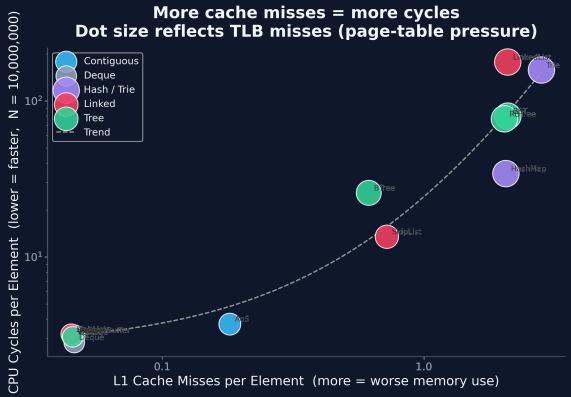
- **Arrays:** visiting element *i* loads *i+1...8* for free.
- **Linked Structures:** every hop is a separate RAM round-trip.
- **The Penalty:** 100+ cycles lost per pointer-follow.

# Empirical Comparison: 16 Structures




Azure = Contiguous (Arrays/Vectors) | Emerald = Trees | Rose = Linked Systems

# The Causality: Misses → Cycles

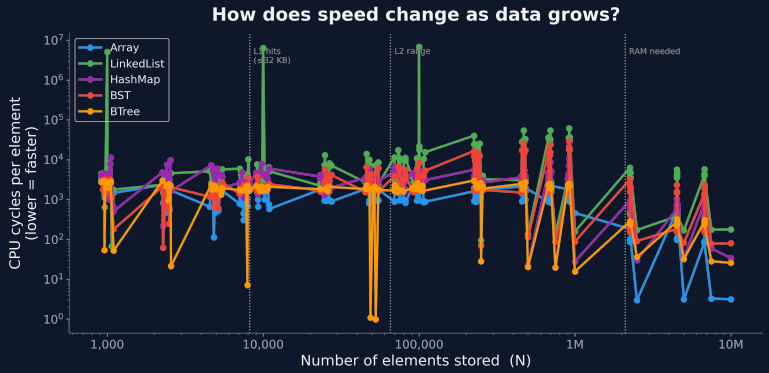



## Causation

Hardware counters confirm the trend.

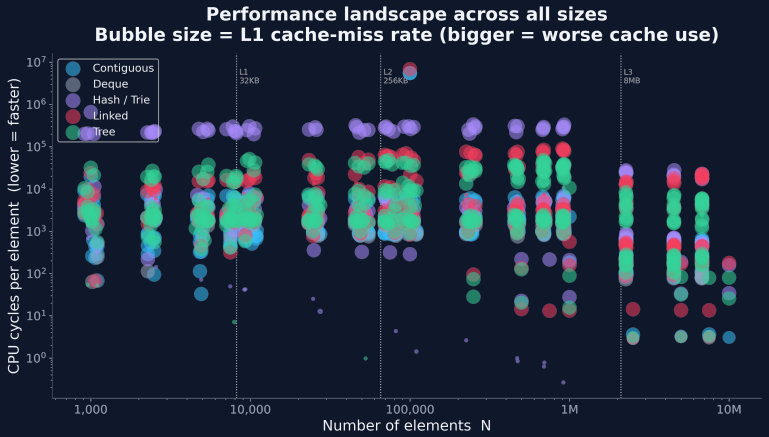
- Cache exhaustion is the primary driver of latency.
- Slope represents cycle per miss penalty.

# Scaling Dynamics




Scaling is non-linear: performance collapses the moment working set > cache size.

# Structural Landscape




The blue cluster (Contiguous) maintains hardware efficiency across 4 orders of magnitude.

# Findings: The Efficiency Hierarchy

- ✔ **Optimal: Arrays, Vectors, SoA** — predictable patterns.
- ↺ **Middling:** BTrees — hybrid structures that batch access.
- ✖ **Suboptimal:** Linked Lists, BSTs — memory latency tax.

## Academic Takeaway

Algorithms optimize operations. **Data representations** optimize hardware.

# Synthesis

**1**
Layout is Logic

**2**
Cache is King

**3**
Pointers are Stalls

The fastest code is the code that doesn't wait on its data.